

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

(Program Elective – IV)

OPTICAL FIBER COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) Assume suitable data wherever necessary.
 - (3) All questions carry marks as indicated.
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1.
 - (a) Define the relative refractive index difference for an optical fiber. Show how it is related to numerical aperture by deriving expression for NA. 3(CO1)
 - (b) A graded index fiber has core with parabolic refractive index profile which has a diameter of $50\text{ }\mu\text{m}$. The fiber has a N.A. of 0.2. Estimate the total number of guided modes propagating in the fiber when it is operating at a wavelength of $1\text{ }\mu\text{m}$. 3(CO1)
 - (c) Differentiate :
 - (i) Meridional rays and Skew rays.
 - (ii) Multimode step index and Multimode graded index fibers. 4(CO1)
 2.
 - (a) Explain the various types of dispersions occurring in optical fibers. Derive the expression for material dispersion parameter. 3(CO2)
 - (b) Explain the losses due to misalignment of optical fibers. 3(CO3)
 - (c) A continuous 12 Km long optical fiber link has 1.5 dB/Km attenuation :
 - (i) What is the minimum optical power level that must be launched in to fiber to maintain optical power level of 0.3 microwatts at the receiving end ?
 - (ii) What is the required input power if fiber has loss of 2.5 dB/Km ? 4(CO3)

3. (a) With the aid of suitable diagram discuss the principle of operation of injection LASER. 5(CO4)
- (b) Estimate the external power efficiency of a planar LED when the transmission factor of LED air interface is 0.68 and internally generated optical power is 30% of electrical power supplied. Refractive index of LED material is 3.6. 5(CO4)
4. (a) Discuss the principle of operation of APD. Discuss advantages and drawbacks. 5(CO4)
- (b) A p-i-n photodiode has quantum efficiency of 0.5 at wavelength of $0.9\mu\text{m}$, Calculate :
 - (i) Received optical power if the mean photocurrent is $1\mu\text{A}$.
 - (ii) The corresponding number of received photons per second at this wavelength. 5(CO4)
5. (a) Discuss with aid of suitable diagram the cutback technique used for the measurement of the total attenuation in an optical fiber. 5(CO5)
- (b) Explain with diagram the process of measurement of Refractive Index Profile of a given Optical fiber. 5(CO5)
6. Write short notes on :—
 - (i) WDM.
 - (ii) SONET.
 - (iii) Link Power Budget.
 - (iv) High speed light wave links. 10(CO5)

